BAHRIA UNIVERSITY KARACHI CAMPUS

DATABASE MANAGEMENT SYSTEM
PROJECT FINAL REPORT

ADVANCED PHARMACY MANAGEMENT SYSTEM

Semester: Spring 2026

Year: CE (6th Semester)

Name	ENROLLMENT NO
Muhammad Arslan	02-132232-009

Course Instructor: **Sidra Mudassar**
Lab Instructor: **Sidra Mudassar**

1. Introduction

The Advanced Pharmacy Management System is a database-driven desktop application designed to manage daily pharmacy operations. The system stores and manages medicines, suppliers, customers, purchases, sales, returns, expenses, users, reports, and audit records.

The project uses Microsoft SQL Server as the main database and Java NetBeans as the frontend application. The frontend communicates with SQL Server through the Microsoft JDBC Driver. All pharmacy records are saved permanently in the SQL Server database.

2. Objectives

- To store pharmacy data in a structured SQL Server database.
- To manage medicines with batch number, barcode, expiry date, quantity, rack number, purchase price, and sale price.
- To manage suppliers, customers, purchases, and sales billing.
- To automatically increase stock after purchase and decrease stock after sale.
- To provide dashboard summaries, stock reports, expiry reports, and profit reports.
- To implement login authentication using users and roles.

3. Scope of the Project

The system supports pharmacy-level inventory and billing operations. It is suitable for an academic database project and demonstrates relational database design, SQL constraints, stored procedures, triggers, views, and Java database connectivity.

The system supports the following users and roles:

- Admin
- Pharmacist
- Cashier
- Manager

4. Tools and Technologies

Component	Technology Used
Database	Microsoft SQL Server
Database Tool	SQL Server Management Studio (SSMS)
Frontend	Java Swing Forms
IDE	Apache NetBeans IDE 29
Database Connectivity	Microsoft JDBC Driver for SQL Server
Programming Language	Java
Database Name	AdvancedPharmacyManagementDB

5. Database Design

The database contains 13 relational tables. Primary keys and foreign keys are used to maintain relationships between the tables.

Table Name	Purpose
Roles	Stores system roles such as Admin, Pharmacist, Cashier, and Manager.
Users	Stores login users and connects each user with a role.
Categories	Stores medicine categories such as Tablet, Syrup, Injection, Capsule, Cream, Drops, and Inhaler.
Suppliers	Stores supplier and company information.
Customers	Stores customer information and contact details.
Medicines	Stores medicine inventory, pricing, batch, barcode, expiry, rack, and stock quantity.
Purchases	Stores purchase header records from suppliers.
PurchaseDetails	Stores medicines included in each purchase.
Sales	Stores sale bill header records.
SaleDetails	Stores medicines included in each sale bill.
Returns	Stores customer, supplier, expired, and damaged returns.
Expenses	Stores pharmacy expenses.

6. Database Relationships

- One Role has many Users.
- One Category has many Medicines.
- One Supplier can supply many Medicines.
- One Supplier can have many Purchases.
- One User can create many Purchases and Sales.
- One Purchase has many PurchaseDetails.
- One Sale has many SaleDetails.
- One Customer can have many Sales.
- One Medicine can appear in many SaleDetails and PurchaseDetails.
- One Medicine can have many Returns.
- One Sale can have many Returns.
- One User can have many AuditLogs.

7. Normalization

The database design follows Third Normal Form (3NF). Data is separated into different tables to avoid duplication and maintain consistency.

- First Normal Form (1NF): Each field contains atomic values and no repeating groups are used.
- Second Normal Form (2NF): Non-key attributes depend on the complete primary key.
- Third Normal Form (3NF): Transitive dependency is avoided by separating roles, categories, suppliers, customers, medicines, sales, and purchases into separate tables.

8. SQL Constraints and Data Integrity

The system uses SQL constraints to protect data accuracy and prevent invalid records.

- PRIMARY KEY is used for unique identification of each record.
- FOREIGN KEY is used to maintain table relationships.
- UNIQUE is used for fields such as username, role name, category name, and barcode.
- NOT NULL is used for required fields.
- CHECK constraints validate status, quantity, price, payment method, and return type.
- Computed columns are used for DueAmount, NetAmount, ChangeAmount, and SubTotal.

9. SQL Views, Stored Procedures, and Triggers

9.1 Views

- ViewAllMedicines
- ViewLowStockMedicines
- ViewOutOfStockMedicines
- ViewExpiredMedicines
- ViewNearExpiryMedicines
- ViewStockReport
- ViewSupplierMedicineReport

9.2 Stored Procedures

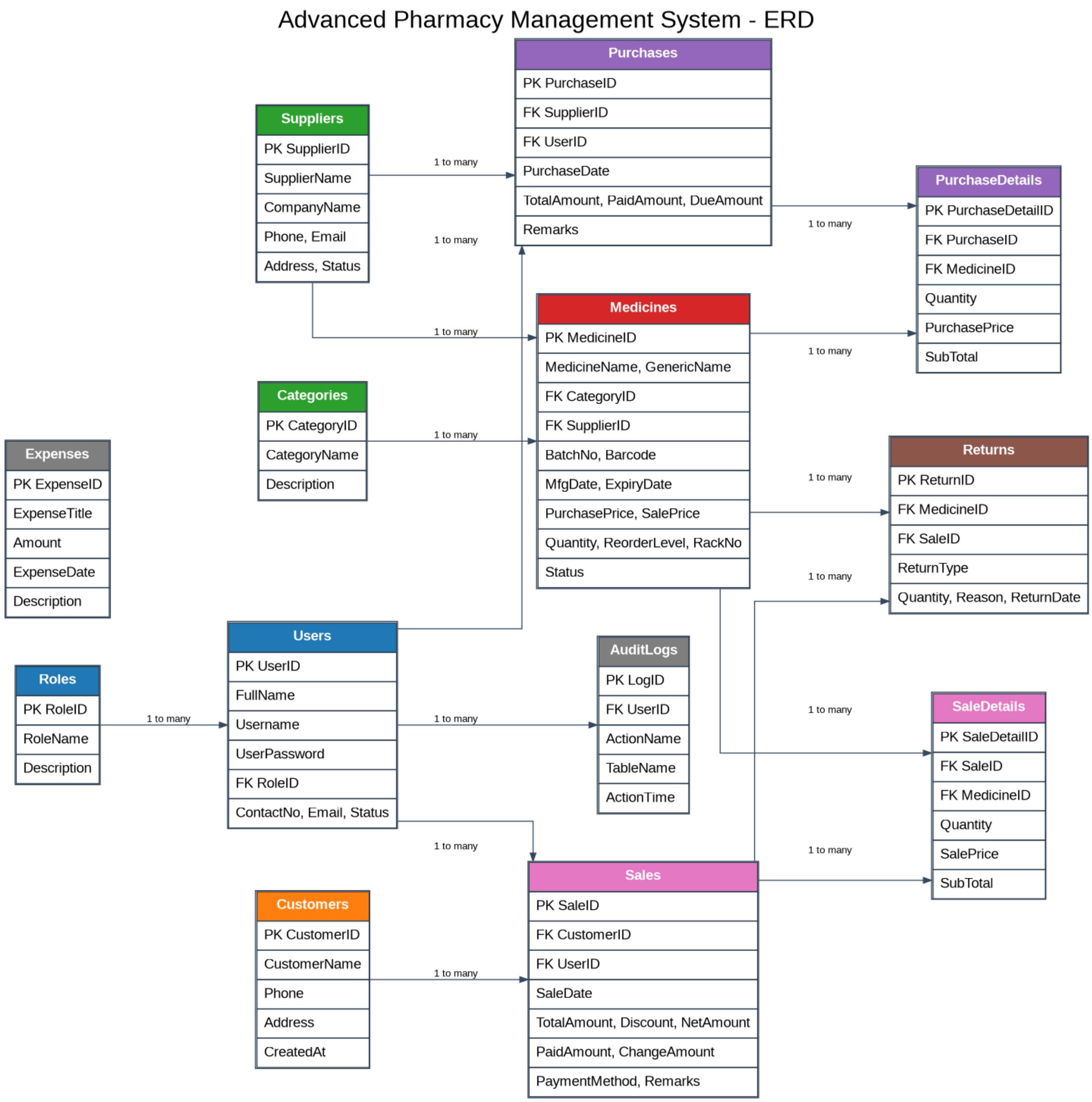
- sp_LoginUser - validates login credentials.
- sp_SearchMedicine - searches medicines by name, generic name, barcode, or batch number.
- sp_AddMedicine, sp_UpdateMedicine, sp_DeleteMedicine - manage medicines.
- sp_CreatePurchase and sp_AddPurchaseDetail - create purchases and purchase items.
- sp_CreateSale and sp_AddSaleDetail - create sales and sale items.
- sp_AddReturn - stores return records.
- sp_DashboardSummary - shows dashboard totals.
- sp_DailySalesReport, sp_MonthlySalesReport, sp_ProfitReport - generate reports.
- sp_CustomerPurchaseHistory - shows customer purchase history.

9.3 Triggers

- trg_IncreaseStockAfterPurchase - increases stock after purchase details are inserted.
- trg_UpdatePurchaseTotal - updates purchase total amount.
- trg_PreventSaleIfStockNotEnough - prevents sale when stock is insufficient.
- trg_UpdateSaleTotal - updates sale total amount.
- trg_LogNewSale - stores sale activity in audit logs.
- trg_LogNewPurchase - stores purchase activity in audit logs.

10. Entity Relationship Diagram (ERD)

The ERD below shows the main entities, primary keys, foreign keys, and relationships used in the Advanced Pharmacy Management System.



11. Application Modules

Module / Form	Function
LoginForm.java	User login using SQL Server Users and Roles tables.
DashboardForm.java	Shows total medicines, suppliers, customers, sales, stock value, low stock, expired, and near-expiry medicines.
MedicineForm.java	Adds, searches, and displays medicines.
SupplierForm.java	Adds, searches, and displays suppliers.
CustomerForm.java	Adds, searches, and displays customers.
PurchaseForm.java	Creates purchases and adds purchase items. Stock increases automatically.
SalesBillingForm.java	Creates sales bills and adds sale items. Stock decreases automatically.
ReportsForm.java	Shows low stock, expired, near expiry, stock, and profit reports.
UserManagementForm.java	Adds and displays system users.
InvoiceForm.java	Displays and prints sale invoice/receipt.

12. SQL Queries Used for Testing

The following SQL queries were used in SQL Server Management Studio to verify the same data stored and displayed by the Java application.

```
USE AdvancedPharmacyManagementDB;
GO

SELECT * FROM ViewAllMedicines;

SELECT * FROM Suppliers;

SELECT * FROM Customers;

SELECT * FROM Purchases;
SELECT * FROM PurchaseDetails;

SELECT * FROM Sales;
SELECT * FROM SaleDetails;

SELECT u.UserID, u.FullName, u.Username, r.RoleName, u.ContactNo, u.Email, u.Status
FROM Users u
INNER JOIN Roles r ON u.RoleID = r.RoleID;

EXEC sp_DashboardSummary;

EXEC sp_ProfitReport;
```

13. User Authentication System

Login credentials are stored in the Users table. The sp_LoginUser stored procedure checks the username and password and returns the user role. The dashboard displays the logged-in user name and role.

- Default Admin Login: admin / admin123
- Default Pharmacist Login: pharmacist / pharma123
- Default Cashier Login: cashier / cashier123
- Default Manager Login: manager / manager123

14. Project Flow

1. User opens the Java application.
2. Login form appears.
3. User enters username and password.
4. SQL Server validates the login through stored procedure sp_LoginUser.
5. Dashboard opens after successful login.
6. User manages medicines, suppliers, customers, purchases, sales, users, invoices, and reports through Java forms.
7. All records are saved in SQL Server.
8. SSMS queries show the same data that appears in the Java application.

15. Conclusion

The Advanced Pharmacy Management System successfully demonstrates the use of SQL Server with a Java NetBeans frontend. The system manages pharmacy inventory, sales, purchases, customers, suppliers, users, invoices, reports, and audit logs. It uses database concepts such as tables, primary keys, foreign keys, constraints, views, stored procedures, triggers, and normalization.

16. Future Enhancements

- Barcode scanner integration.
- Advanced role-based screen restrictions.
- Medicine return invoice printing.
- Automatic expiry notification alerts.
- Graphical sales dashboard.
- Cloud database hosting.